60V PNP MEDIUM POWER TRANSISTOR IN SOT223

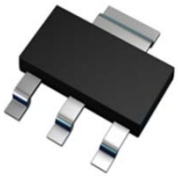
Features

- BV_{CEO} > -60V
- I_C = -5A High Continuous Collector Current
- I_{CM} = -15A Peak Pulse Current
- Low Saturation Voltage $V_{CE(SAT)}$ < -140mV @ -1A
- $R_{CE(SAT)}$ = 55mΩ for a Low Equivalent On-Resistance
- h_{FE} Specified up to -10A for a High Gain Hold-Up
- Complementary NPN Type: FZT851
- **Lead-Free Finish; RoHS Compliant (Notes 1 & 2)**
- **Halogen and Antimony Free. "Green" Device (Note 3)**
- **Qualified to AEC-Q101 Standards for High Reliability**
- **PPAP Capable (Note 4)**

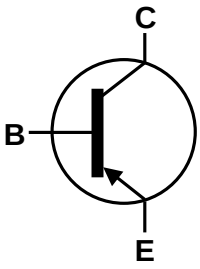
Mechanical Data

- Case: SOT223
- Case Material: Molded Plastic. "Green" Molding Compound. UL Flammability Rating 94V-0
- Moisture Sensitivity: Level 1 per J-STD-020
- Terminals: Finish—Matte Tin Plated Leads, Solderable per MIL-STD-202, Method 208 e3
- Weight: 0.112 grams (Approximate)

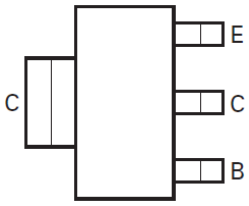
SOT223



Top View



Device Symbol



Top View
Pin-Out

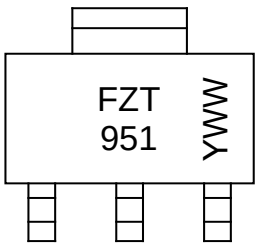
Ordering Information (Notes 4 & 5)

Product	Compliance	Marking	Reel Size (inches)	Tape Width (mm)	Quantity per Reel
FZT951TA	AEC-Q101	FZT951	7	12	1000
FZT951TC	AEC-Q101	FZT951	13	12	4000
FZT951QTA	Automotive	FZT951	7	12	1000
FZT951QTC	Automotive	FZT951	13	12	4000

Notes:
1. EU Directive 2002/95/EC (RoHS), 2011/65/EU (RoHS 2) & 2015/863/EU (RoHS 3) compliant. All applicable RoHS exemptions applied.
2. See https://www.diodes.com/quality/lead-free/ for more information about Diodes Incorporated's definitions of Halogen- and Antimony-free, "Green" and Lead-free.
3. Halogen- and Antimony-free "Green" products are defined as those which contain <900ppm bromine, <900ppm chlorine (<1500ppm total Br + Cl) and <1000ppm antimony compounds.
4. Automotive products are AEC-Q101 qualified and are PPAP capable. Refer to https://www.diodes.com/quality/.
5. For packaging details, go to our website at https://www.diodes.com/design/support/packaging/diodes-packaging/.

Marking Information

SOT223



FZT 951 = Product Type Marking Code
YWW = Date Code Marking
Y or $\overline{Y}$ = Last Digit of Year (ex: 8 = 2018)
WW or $\overline{WW}$ = Week Code (01–53)

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Absolute Maximum Ratings (@T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Value	Unit
Collector-Base Voltage	V_{CBO}	-100	V
Collector-Emitter Voltage	V_{CEO}	-60	V
Emitter-Base Voltage	V_{EBO}	-7	V
Continuous Collector Current	I_C	-5	A
Peak Pulse Current	I_{CM}	-15	A

Thermal Characteristics (@T_A = +25°C, unless otherwise specified.)

Characteristic		Symbol	Value	Unit
Power Dissipation Linear Derating Factor	(Note 6)	P_D	3.0 24	W mW /°C
	(Note 7)		1.6 12.8	
Thermal Resistance, Junction to Ambient	(Note 6)	$R_{\Theta JA}$	42	°C/W
	(Note 7)	$R_{\Theta JA}$	78	
Thermal Resistance Junction to Lead	(Note 8)	$R_{\Theta JL}$	8.8	
Operating and Storage Temperature Range		T_J, T_{STG}	-55 to +150	°C

ESD Ratings (Note 9)

Characteristic	Symbol	Value	Unit	JEDEC Class
Electrostatic Discharge—Human Body Model	ESD HBM	8000	V	3B
Electrostatic Discharge—Machine Model	ESD MM	400	V	C

Notes:
6. For a device mounted with the collector lead on 52mm × 52mm 2oz copper that is on a single-sided 1.6mm FR4 PCB; device is measured under still air conditions whilst operating in steady-state.
7. Same as Note 6, except the device is mounted on 25mm × 25mm 1oz copper.
8. Thermal resistance from junction to solder-point (at the end of the collector lead).
9. Refer to JEDEC specification JESD22-A114 and JESD22-A115.

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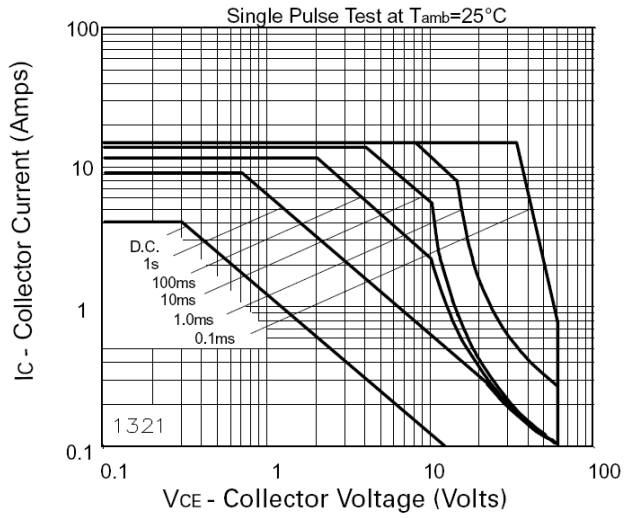
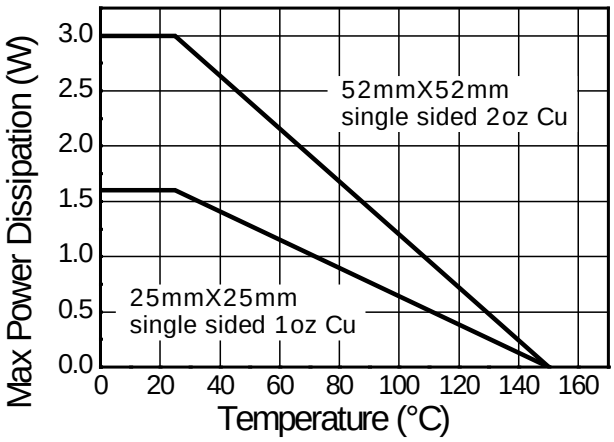
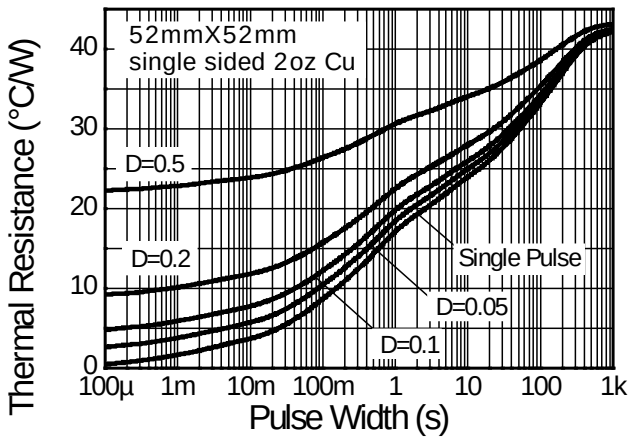
Thermal Characteristics and Derating Information



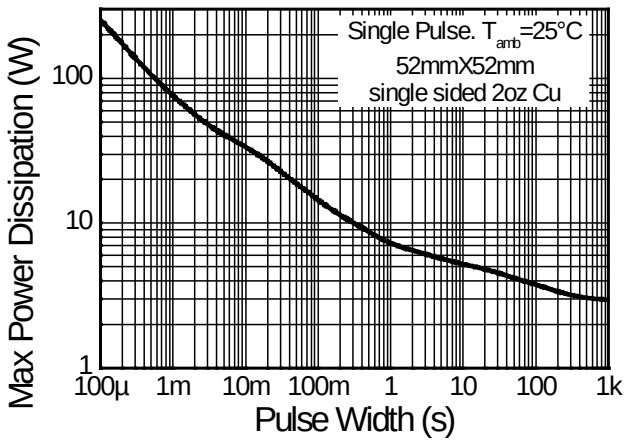
Safe Operating Area



Derating Curve



Transient Thermal Impedance



Pulse Power Dissipation

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Electrical Characteristics (@T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Min	Typ.	Max	Unit	Test Condition
Collector-Base Breakdown Voltage	BV_{CBO}	-100	-140	—	V	I_C = -100µA
Collector-Emitter Breakdown Voltage (Note 10)	BV_{CER}	-100	-140	—	V	I_C = -1µA, R_B ≤ 1kΩ
Collector-Emitter Breakdown Voltage (Note 10)	BV_{CEO}	-60	-90	—	V	I_C = -10mA
Emitter-Base Breakdown Voltage	BV_{EBO}	-7	-8	—	V	I_E = -100µA
Collector Cut-Off Current	I_{CBO}	—	<-1 —	-50 -1	nA µA	V_{CB} = -80V V_{CB} = -80V, T_A = +100°C
Collector Cut-Off Current	I_{CER}	—	<-1 —	-50 -1	nA µA	V_{CE} = -80V, R ≤ 1kΩ V_{CE} = -80V, T_A = +100°C
Emitter Cut-Off Current	I_{EBO}	—	<1	-10	nA	V_{EB} = -6V
DC Current Transfer Static Ratio (Note 10)	h_{FE}	100	200	—	—	I_C = -10mA, V_{CE} = -1V
		100	200	300		I_C = -2A, V_{CE} = -1V
		75	90	—		I_C = -5A, V_{CE} = -1V
		10	25	—		I_C = -10A, V_{CE} = -1V
Collector-Emitter Saturation Voltage (Note 10)	$V_{CE(SAT)}$	—	-20	-50	mV	I_C = -100mA, I_B = -10mA
		—	-85	-140		I_C = -1A, I_B = -100mA
		—	-155	-210		I_C = -2A, I_B = -200mA
		—	-370	-460		I_C = -5A, I_B = -500mA
Base-Emitter Saturation Voltage (Note 10)	$V_{BE(SAT)}$	—	-1080	-1240	mV	I_C = -5A, I_B = -500mA
Base-Emitter Turn-On Voltage (Note 10)	$V_{BE(ON)}$	—	-935	-1070	mV	I_C = -5A, V_{CE} = -1V
Transitional Frequency (Note 10)	f_T	—	120	—	MHz	I_C = -100mA, V_{CE} = -10V, f = 50MHz
Output Capacitance	C_{OBO}	—	74	—	pF	V_{CB} = -10V, f = 1MHz
Switching Time	t_{ON}	—	82	—	ns	V_{CC} = -10V, I_C = -2A, $-I_{B1}$ = I_{B2} = -200mA
	t_{OFF}	—	350	—		

Note: 10. Measured under pulsed conditions. Pulse width ≤ 300µs. Duty cycle ≤ 2%.

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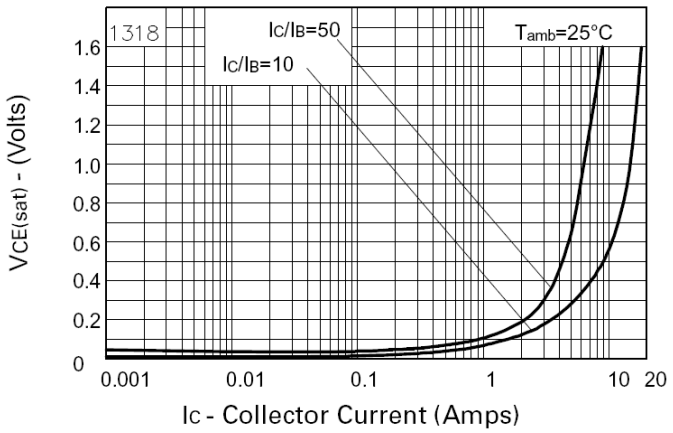
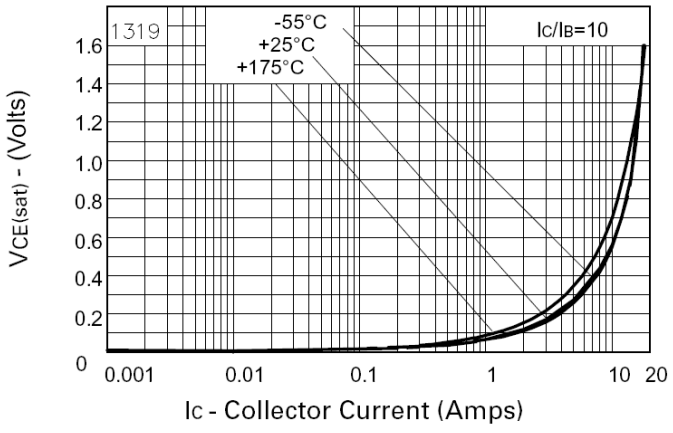
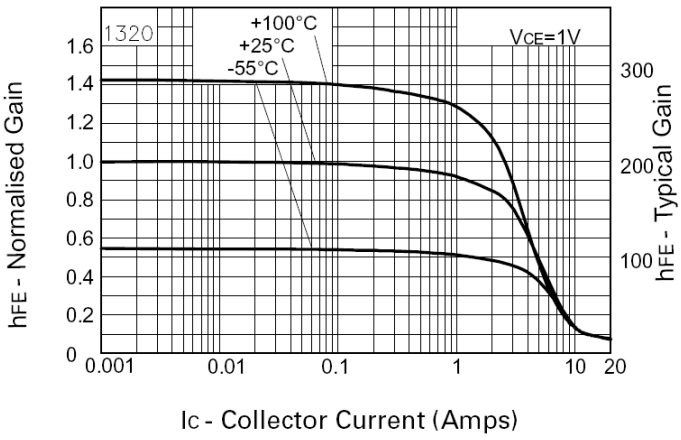
Typical Electrical Characteristics (@T_A = +25°C, unless otherwise specified.)



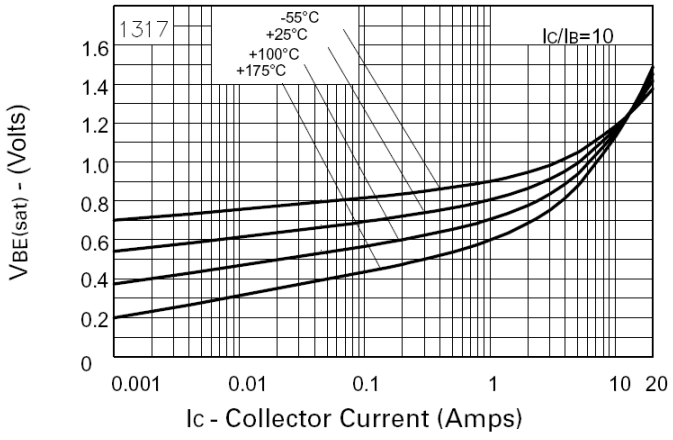
VCE(sat) v IC



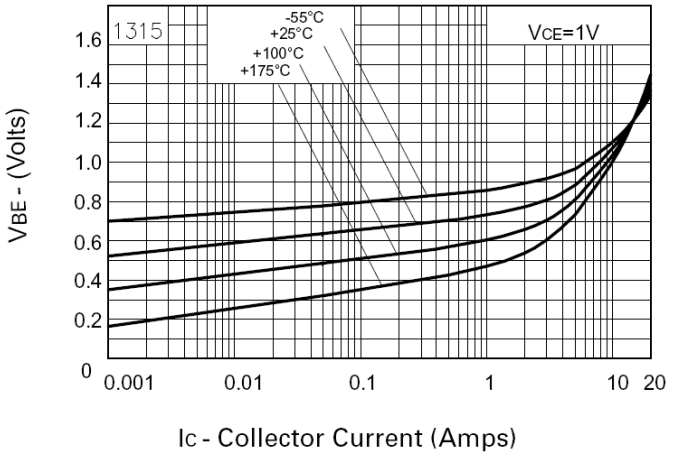
VCE(sat) v IC



hFE v IC



VBE(sat) v IC



VBE(on) v IC

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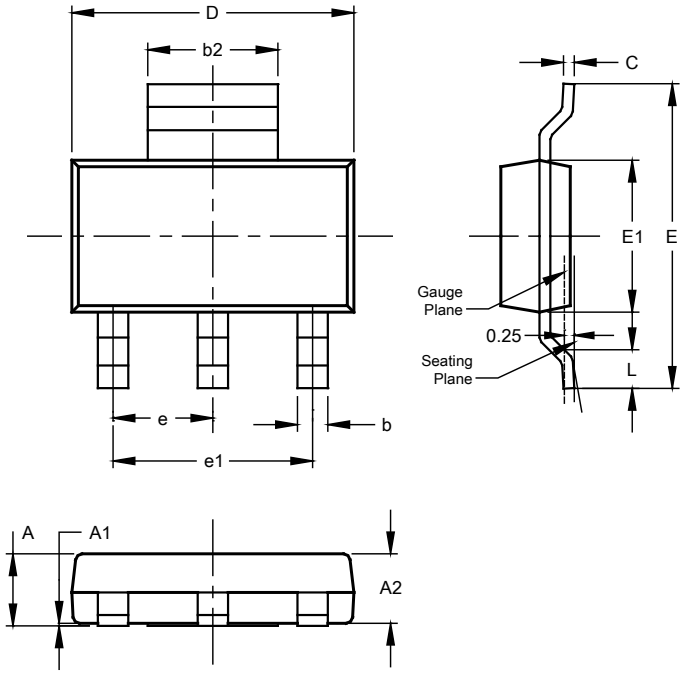
Package Outline Dimensions

Please see http://www.diodes.com/package-outlines.html for the latest version.

SOT223 (Type DN)

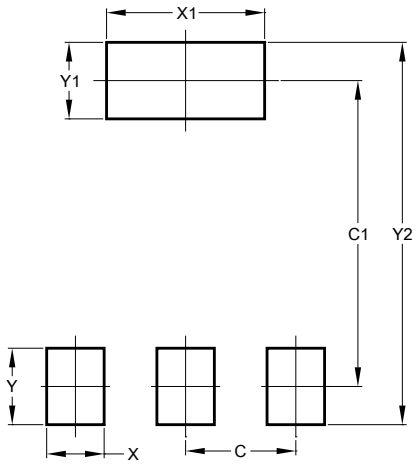


SOT223 (Type DN)			
Dim	**Min**	**Max**	**Typ**
A	--	1.70	--
A1	0.01	0.15	--
A2	1.50	1.68	1.60
b	0.60	0.80	0.70
b2	2.90	3.10	--
c	0.20	0.32	--
D	6.30	6.70	--
E	6.70	7.30	--
E1	3.30	3.70	--
e	--	--	2.30
e1	--	--	4.60
L	0.85	--	--
All Dimensions in mm			

Suggested Pad Layout

Please see http://www.diodes.com/package-outlines.html for the latest version.

SOT223 (Type DN)



Dimensions	Value (in mm)
C	2.30
C1	6.40
X	1.20
X1	3.30
Y	1.60
Y1	1.60
Y2	8.00

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